Kodewala Academy – Java Backend Tests

Complete Question Bank · Test 1 · 39 Questions

Test 1 – Java Backend Test #1

Q1. Which of the following is a valid variable name?

A. 1name

B. class

C. studentName

D. student-name

Correct answer: C. studentName

Explanation: Variable names cannot start with numbers, contain hyphens, or use reserved keywords.

Q2. Which command compiles a Java program?

A. java Test

B. compile Test.java

C. javac Test

D. javac Test.java

Correct answer: D. javac Test.java

Explanation: javac converts source code into bytecode.

Q3. Which symbol is used to terminate a statement in Java?

A. :

B. ,

C. ;

D. .

Correct answer: C. ;

Explanation: Most Java statements end with a semicolon.

Q4. Which component executes Java bytecode?

A. JDK

B. JVM

C. JRE

D. Compiler

Correct answer: B. JVM

Explanation: JVM loads and executes .class files.

Q5. Which variable belongs to an object?

A. Static Variable

B. Local Variable

C. Final Variable

D. Instance Variable

Correct answer: D. Instance Variable

Explanation: Every object gets its own copy of instance variables.

Q6. Which variable is shared among all objects?

A. Static Variable

B. Local Variable

C. Final Variable

D. Instance Variable

Correct answer: A. Static Variable

Explanation: Static variables belong to the class, not objects.

Q7. What is the output?

```
System.out.println(10 + 20 + "Java");
```

A. Java1020

B. 1020Java

C. 30Java

D. Error

Correct answer: C. 30Java

Explanation: Integer addition happens before string concatenation.

Q8. Which statement is correct?

A. JVM contains JRE

B. JRE contains JVM

C. JDK contains only JVM

D. JRE contains JDK

Correct answer: B. JRE contains JVM

Explanation: JRE = JVM + runtime libraries.

Q9. What is the output?

```
System.out.println(10 + "20" + 30);
```

A. 102030

B. 60

C. Error

D. 3030

Correct answer: A. 102030

Explanation: Once a String appears, concatenation starts.

Q10. What is the output?

```
byte b = 10;
byte c = 20;
byte d = b + c;
```

A. 0

B. 30

C. Compilation Error

D. Runtime Error

Correct answer: C. Compilation Error

Explanation: b + c becomes int, not byte.

Q11. What is the output?

```
String[] arr = {"Java", "Python"};
System.out.println(arr.length);
```

A. 1

B. 2

C. Error

D. 0

Correct answer: B. 2

Explanation: Array contains two elements.

Q12. Is the main() method mandatory in every Java class?

A. Yes, every class must have a main method

B. No, only the class that needs to be executed directly requires it

C. No class can have a main method

D. Every object must have a main method

Correct answer: B. No, only the class that needs to be executed directly requires it

Explanation: A Java class can exist without a main() method. However, if you want to run the class directly using the JVM (java ClassName), it must contain a valid main() method.

Q13. Which is the correct signature of the main() method?

A. public static void main(String args)

B. public void static main(String[] args)

C. public static void main(String[] args)

D. static public main(String[] args)

Correct answer: C. public static void main(String[] args)

Explanation: The JVM looks for exactly public static void main(String[] args).

Q14. Which of the following is a valid Java class name?

A. 123Student

B. student-details

C. StudentDetails

D. class

Correct answer: C. StudentDetails

Explanation: Class names follow identifier rules: they cannot start with a digit, contain hyphens, or be a reserved keyword. StudentDetails also follows the PascalCase convention.

Q15. What happens if a class does not contain a main() method and you try to run it?

A. Compiles and runs successfully

B. Runtime Error: Main method not found

C. JVM creates a main method automatically

D. Compilation Error

Correct answer: B. Runtime Error: Main method not found

Explanation: The class will compile successfully, but when executed, the JVM reports that the main method was not found.

Q16. What is the primary purpose of JIT (Just-In-Time) Compiler?

A. To compile .java files into .class files

B. To create JVM

C. To load classes into memory

D. To convert bytecode into machine code during execution for better performance

Correct answer: D. To convert bytecode into machine code during execution for better performance

Explanation: JIT is a part of the JVM. It converts frequently executed bytecode into native machine code, making Java programs run faster.

Note: This question appears twice (Q16 and Q17) in the supplied test with identical content.

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Q18. Why should hardcoding values be avoided in Java applications?

A. It improves performance

B. It reduces memory usage

C. It makes maintenance and configuration difficult

D. It is required by the JVM

Correct answer: C. It makes maintenance and configuration difficult

Explanation: Hardcoded values (database URLs, passwords, file paths, etc.) make applications difficult to maintain. Such values should be stored in configuration files, environment variables, or properties files.

Q19. Why is Java called a Platform Independent language?

A. Java programs do not require a compiler

B. Java source code runs directly on the operating system

C. Java bytecode can run on any platform that has a JVM

D. Java works only on Windows and Linux

Correct answer: C. Java bytecode can run on any platform that has a JVM

Explanation: When Java code is compiled, it is converted into bytecode (.class file). This bytecode is not specific to any operating system. Any machine with a JVM can execute it. This is the reason behind Java's principle: "Write Once, Run Anywhere (WORA)".

Q20. Where are objects stored in Java memory?

A. Stack Memory

B. Heap Memory

C. Method Area

D. CPU Cache

Correct answer: B. Heap Memory

Explanation: Objects created using the new keyword are stored in Heap Memory. The reference variable may be stored elsewhere, but the actual object resides in the heap.

Q21. Where are local variables stored?

A. Heap Memory

B. MetaSpace

C. PermGen

D. Stack Memory

Correct answer: D. Stack Memory

Explanation: Local variables declared inside methods are commonly described as being stored in stack frames and are no longer available when the method execution completes.

Q22. What is a common use case of a static variable in Java?

A. Storing object-specific data

B. Storing data shared by all objects of a class

C. Creating objects

D. Storing local variables

Correct answer: B. Storing data shared by all objects of a class

Explanation: Static variables belong to the class, not to individual objects. One class-level copy is shared by all objects.

Q23. What is the scope of a local variable in Java?

A. Entire class

B. Entire package

C. Only within the method/block where it is declared

D. Entire application

Correct answer: C. Only within the method/block where it is declared

Explanation: A local variable can be accessed only inside the method, constructor, or block in which it is declared. It cannot be accessed outside that scope.

Q24. Which statement about the main() method is TRUE?

A. It must always be the first method in the class.

B. It can be non-static.

C. A class can compile without a main method.

D. Every class must contain a main method.

Correct answer: C. A class can compile without a main method.

Explanation: A class without main() compiles successfully. The JVM requires main() only when you try to run that class directly.

Q25. What is the relationship between a .java file, classes, and .class files?

Consider the following code saved in Test.java:

```
class A {
}
class B {
}
public class Test {
    public static void main(String[] args) {
    }
}
```

After compilation (javac Test.java), how many .class files will be generated?

A. Compilation Error

B. 1

C. 2

D. 3

Correct answer: D. 3

Explanation: Java creates one .class file for each top-level class in this example.

Q26. Which statement is TRUE?

A. One .java file can contain only one class.

B. One .java file can contain multiple classes.

C. One class generates multiple .class files always.

D. A .class file contains Java source code.

Correct answer: B. One .java file can contain multiple classes.

Explanation: A Java source file can contain multiple classes, but at most one public top-level class. The file name must match the public class name.

Q27. File Name is Demo.java. What will happen?

```
public class Test {
    public static void main(String[] args) {
        System.out.println("Hello");
    }
}
```

A. Runs successfully

B. Generates Demo.class

C. Compilation Error

D. Runtime Error

Correct answer: C. Compilation Error

Explanation: When a class is declared public, the file name must match the public class name.

Q28. What will happen when the following program is executed without any command-line arguments?

```
public class Test {
    public static void main(String[] args) {
        System.out.println(args[0]);
    }
}
```

A. null

B. blank output

C. ArrayIndexOutOfBoundsException

D. Compilation failure

Correct answer: C. ArrayIndexOutOfBoundsException

Explanation: Since no arguments are passed, args.length is 0. Accessing args[0] causes an ArrayIndexOutOfBoundsException at runtime.

Q29. What is the output?

```
public class Test {
    public static void main(String[] args) {
        if (args.length > 0) {
            System.out.println(args[0]);
        } else {
            System.out.println("No Arguments");
        }
    }
}
```

Executed as: java Test

A. null

B. No Arguments

C. ArrayIndexOutOfBoundsException

D. Compilation failure

Correct answer: B. No Arguments

Explanation: The code checks args.length before accessing the array, preventing ArrayIndexOutOfBoundsException.

Q30. What is the output?

```
public class Test {
    public static void main(String[] args) {
        System.out.println(args.length);
    }
}
```

Executed as: java Test Java Spring Boot

A. 2

B. 3

C. 4

D. Compilation failure

Correct answer: B. 3

Explanation: The array contains three elements: Java, Spring, and Boot.

Q31. A Java Employee class is responsible for:

```
class Employee {
    void calculateSalary() {
        // salary calculation
    }
    void generateSalarySlip() {
        // generate PDF
    }
    void sendEmail() {
        // send salary slip by email
    }
}
```

Which SOLID principle is violated in this design?

A. Open/Closed Principle

B. Single Responsibility Principle

C. Liskov Substitution Principle

D. Dependency Inversion Principle

Correct answer: B. Single Responsibility Principle

Explanation: The class mixes salary calculation, PDF generation, and emailing. Each class should have only one reason to change, so this violates the Single Responsibility Principle.

Q32. What does the “S” in SOLID stand for?

A. Service Segregation Principle

B. Single Responsibility Principle

C. System Responsibility Principle

D. Software Responsibility Principle

Correct answer: B. Single Responsibility Principle

Explanation: SOLID = Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion.

Q33. A developer writes the following code:

```
class PaymentService {
    void processPayment() {
        String paymentUrl = "https://payment.example.com/api";
        // payment logic
    }
}
```

The payment URL is hardcoded inside the Java class. What is the main disadvantage of hardcoding values like this?

A. It makes the application faster

B. It makes the code easier to maintain and change

C. Changes require modifying and redeploying the source code

D. It automatically improves security

Correct answer: C. Changes require modifying and redeploying the source code

Explanation: A hardcoded value is baked into the source, so a change (such as a different URL per environment) needs a code change and redeployment. Externalize it in properties files or environment variables.

Q34. Consider the following code:

```
class Test {
    static void display() {
        System.out.println("Hello");
    }
    public static void main(String[] args) {
        display();
    }
}
```

When display() is called, where is the method call information stored during execution?

A. Heap

B. Stack

C. String Pool

D. CPU Cache

Correct answer: B. Stack

Explanation: Each method call creates a stack frame holding method execution information, such as local variables and return information. The frame is removed when the method returns.

Q35.

```
int a = 10;
int b = 20;
System.out.println(a > 5 && b < 15);
```

What is the output?

A. true

B. false

C. Runtime error

D. Compilation error

Correct answer: B. false

Explanation: a > 5 is true but b < 15 is false (20 < 15), so true && false = false.

Q36.

```
int a = 10;
if (a > 5 || ++a > 10) {
    System.out.println(a);
}
```

A. 10

B. 11

C. Runtime error

D. Compilation error

Correct answer: A. 10

Explanation: a > 5 is already true, so || short-circuits and ++a is never evaluated. a stays 10.

Q37. A software company starts a new project. Before writing any code, the team meets with the client to understand business requirements, features, and user expectations. Which phase of the Software Development Life Cycle (SDLC) is this?

A. Design

B. Testing

C. Implementation (Development)

D. Requirement Analysis

Correct answer: D. Requirement Analysis

Explanation: Gathering business requirements and client expectations before coding is the Requirement Analysis phase of the SDLC.

Q38. What is a JAR file in Java?

A. Java Application Runtime file

B. Java Archive file used to package Java classes and resources

C. Java Assembly Reference file

D. Java Application Registry file

Correct answer: B. Java Archive file used to package Java classes and resources

Explanation: A JAR (Java Archive) bundles compiled classes and resources into a single distributable file.

Q39. Which command is used to create a JAR file?

A. jar cvf app.jar *.class

B. jar xvf app.jar

C. javac app.jar

D. java create app.jar

Correct answer: A. jar cvf app.jar *.class

*Explanation: jar cvf app.jar *.class: c = create, v = verbose, f = output file name.*